MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 6N7 AND 6N7GT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tubes described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Twin triode, medium Mu

Outline
- 6N7 --- B-6 (EIA)
- 6N7GT --- 9-11 or 9-41 (EIA)

Base
- 6N7 --- 99-21
- 6N7GT --- B8-6 or B8-46

Envelope
- 6N7 --- ST9
- 6N7GT --- T9

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element									
6N7	---	sh	h	2a	2g	1g	1a	h	k
6N7GT	---	nc	h	2a	2g	1g	1a	h	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	ib	Ik/k	Pp/a	Ehk	Alt
Unit:	V	V dc	V dc	ma	mA dc	W	v	ft
Maximum:	6.9	330	---	140	55	6	100	10,000
Minimum:	5.7	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	300	-6	---	---	---	---	---

GENERAL:

(B) Qualification - Not required

(9) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	1					
1266	Total grid current	2,3	Ec = -2 V dc	Ic	0	-2.0	uA dc
1256	Electrode current (2) (anode)	4	Ec = 0	Ib	12	23	mA dc
1341	Power output	3,4	Ec = -5 V dc; Esig = 35 V ac; Rp = 2,500 ohms; Rg = 500 ohms	Po	4.2	---	W
1246	Audio frequency noise	2	Esig = 70 mV ac; Rp = 10,000 ohms	EB	---	17	vu
1231	Emission	4	Eb = Ec = 50 V dc	Is	125	---	mA dc
1201	Short and discontinuity detection			---	---	---	---
	Quality conformance inspection, part 2						
1031	Low-frequency vibration	2	Ebb = 250 V dc; Rp = 10,000 ohms	Ep	---	400	mV ac
1301	Heater current			If	740	860	mA
1336	Heater-cathode leakage			Ihk	---	30	uA dc
1256	Electrode current (1) (anode)	2		Ib	5.0	9.0	mA dc
1306	Transconductance	2		Sm	2,725	3,700	umhos
1316	Amplification factor	2		Mu	30	40	---
1101	Secureness of base, cap, or insert			---	---	---	---
1111	Base pin solder depth			---	---	---	---
1211	Insulation of electrodes	5		---	---	---	---
1105	Permanence of marking			---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	2	Group A; Ec = 0; Ehk = 100 V	---	---	---	---
---	Life-test end point (500 hours)						
1341	Power output	4	Ec = -5 V dc; Esig = 35 V ac; Rp = 2,500 ohms; Rg = 500 ohms	Po	2.8	---	W
(B) 1216	Base material insulating quality	6		---	---	---	---

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NOTES:

1. The AQL (percent defective) for each test listed under QCI, part 1, shall be 0.65, inspection level II.
2. Tie 1k to 2k; 1g to 2g; and 1a to 2a at socket.
3. This test shall be performed at the conclusion of the holding period.
4. Test each unit separately.
5. The AQL (percent defective) shall be 4.0, inspection level S3.
3.6. This test shall be performed during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, the test shall be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - ER
Navy - EC
Air Force - 99

Review activities:
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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